

Independent Study: Stochastic Methods in Quantitative Finance
Project 1 — Statistical Arbitrage via Ornstein–Uhlenbeck Mean Reversion

Dante Buhl | Self-paced (target: 4–5 weeks)

Project Description

Two assets whose prices share a long-run equilibrium (a *cointegrated pair*) produce a spread that wanders away from its mean and is pulled back. This project models that spread as an Ornstein–Uhlenbeck (OU) process,

$$dX_t = \theta(\mu - X_t) dt + \sigma dW_t, \quad \theta, \sigma > 0, \quad (1)$$

estimates its parameters from market data, and builds a trading rule that bets on reversion. The goal is *not* a high Sharpe ratio; it is a rigorous, honest study: correct estimation, out-of-sample testing, realistic transaction costs, and a clear account of when the model fails.

Relevant Coursework and Reading

- **Primary text:** A. Papanicolaou, *Introduction to Stochastic Differential Equations (SDEs) for Finance*, New York University course notes (arXiv:1504.05309). Review the material on Brownian motion, the Itô integral and Itô’s lemma, and linear SDEs before Problem 1. (*Add chapter/section numbers from your copy here.*)
- **AM 216 — Stochastic Differential Equations:** Itô calculus, OU process, Euler–Maruyama.
- **AM 213B — Numerical Methods for DEs:** strong/weak convergence of SDE schemes (Problem 2).
- **AM 213A — Numerical Linear Algebra:** least squares and conditioning (Problems 2–4).
- **STAT 131 — Probability Theory:** Gaussian processes, maximum likelihood.

Note: the Papanicolaou notes focus on option pricing, so they supply the stochastic-calculus foundation; cointegration testing and filtering (Problems 3–4) go beyond them.

Problem 1: The OU Process (Theory)

(a) Apply Itô’s lemma to $e^{\theta t} X_t$ to solve (1):

$$X_t = \mu + (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s.$$

(b) Show that $X_t \mid X_0$ is Gaussian with $\mathbb{E}[X_t \mid X_0] = \mu + (X_0 - \mu)e^{-\theta t}$ and $\text{Var}[X_t \mid X_0] = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t})$. Identify the stationary distribution.

- (c) Define the *half-life* of a deviation and show it equals $\ln 2/\theta$. Interpret it as a natural holding period.
- (d) **Exact discretization.** For a sampling interval Δt , show that $X_{k+1} = a + bX_k + \varepsilon_k$ with $b = e^{-\theta\Delta t}$, $a = \mu(1 - b)$, and $\varepsilon_k \sim \mathcal{N}\left(0, \frac{\sigma^2}{2\theta}(1 - b^2)\right)$ i.i.d. Invert these relations to express (θ, μ, σ) in terms of $(a, b, \text{Var } \varepsilon)$.

Problem 2: Simulation and Estimation

- (a) Simulate (1) with Euler–Maruyama and with the exact scheme from 1(d). Measure the strong error of Euler–Maruyama against step size and confirm its expected order of convergence.
- (b) Using 1(d), show that maximum-likelihood estimation of (θ, μ, σ) reduces to least-squares regression of X_{k+1} on X_k . Implement it.
- (c) On synthetic paths with known parameters, study estimator bias and variance as the sample length shrinks. In particular, quantify how poorly θ (and hence the half-life) is identified on short windows. This sets expectations for real data.

Problem 3: Pair Selection and the Spread

- (a) Choose a small universe (e.g. one sector’s large-cap stocks or ETFs) and download daily prices. Split the data into a *formation* period and a later, untouched *test* period.
- (b) In the formation period, test candidate pairs for cointegration (Engle–Granger), estimate the hedge ratio β by regression, and form the spread $S_t = \log P_t^A - \beta \log P_t^B$.
- (c) Fit the OU model to each spread (Problem 2) and keep pairs whose half-life is short relative to the test period.
- (d) Testing many pairs guarantees false positives. Discuss how many pairs you tested and how you account for multiple testing.

Problem 4: Time-Varying Hedge Ratio

- (a) Write the state-space model $\log P_t^A = \alpha_t + \beta_t \log P_t^B + v_t$, with (α_t, β_t) evolving as a random walk. Derive the Kalman filter recursions.
- (b) Compare the static- β and Kalman- β spreads: is the Kalman spread closer to stationary? Does it improve out-of-sample results, or just fit noise?

Problem 5: Trading Rule and Backtest

- (a) Trade the standardized spread $z_t = (S_t - \hat{\mu})/\hat{\sigma}_{\text{stat}}$: enter at $|z_t| > z_{\text{in}}$, exit at $|z_t| < z_{\text{out}}$. Use only information available at time t (no lookahead).

- (b) Run the rule on the test period only. Report Sharpe ratio, maximum drawdown, turnover, and number of trades, both *before and after* transaction costs. Show how performance degrades as costs rise.
- (c) Identify at least one period where a pair’s relationship broke down. Explain what happened and whether any diagnostic (e.g. a drifting half-life) could have warned you.

Deliverables

- A short written report (6 pages max) with derivations for Problem 1 and results for Problems 2–5.
- Clean, documented code (public repository) that reproduces every figure and table.
- A one-paragraph “limitations” section: why you do or do not trust the results.

Optional Extensions

Parallelize the pair search and parameter estimation across a large universe and report the speedup (connects to AM 250). Derive entry/exit thresholds from OU first-passage times rather than choosing them by hand.